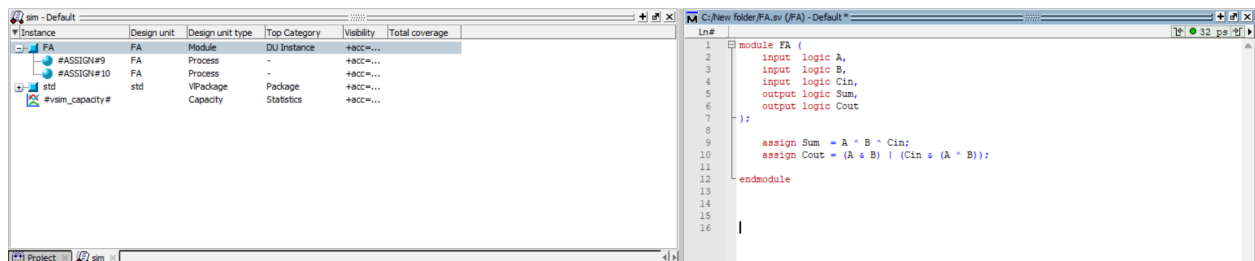


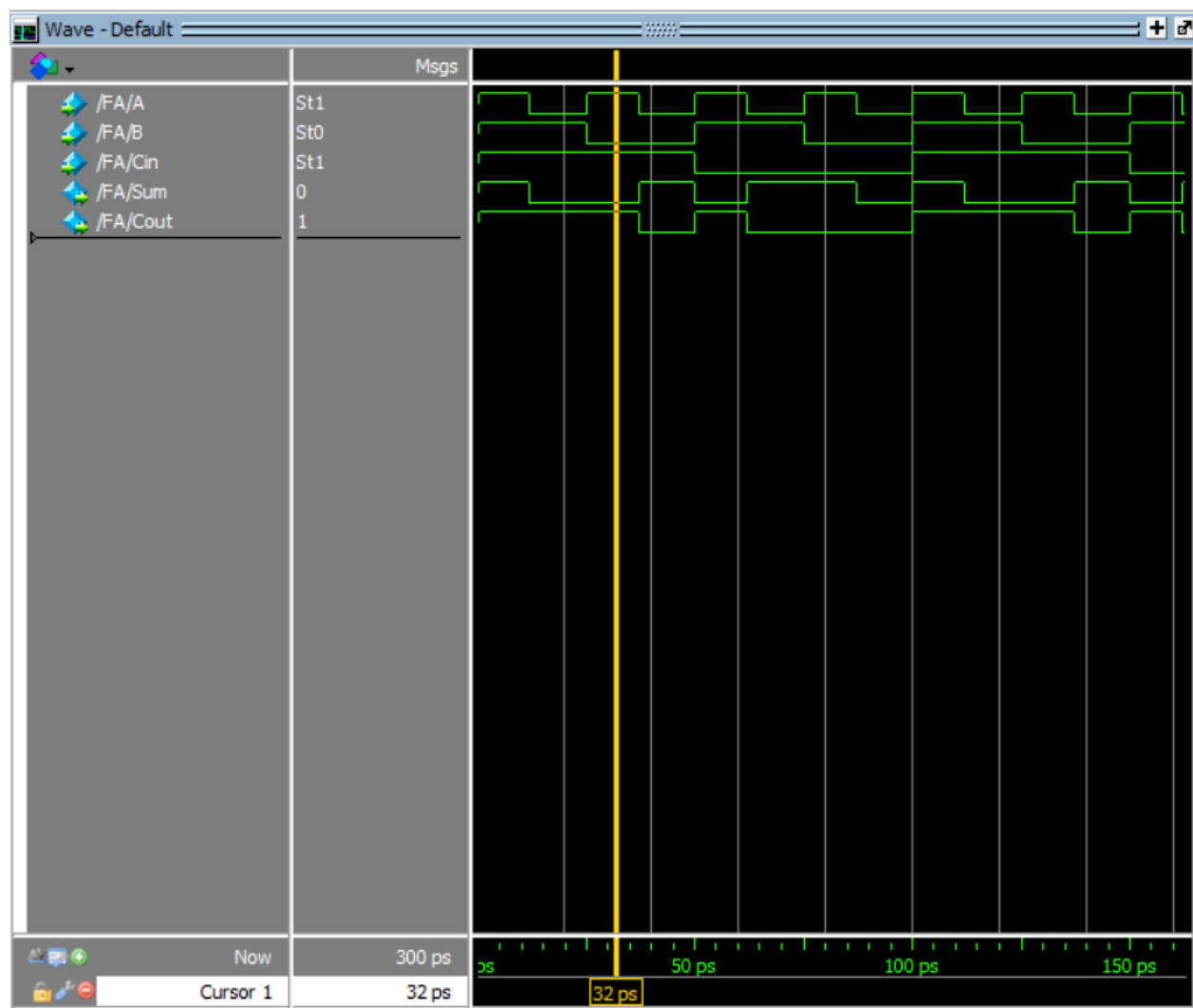
Truth table:

A	B	Cin	Sum	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Code:



Wave form:



Netlist RTL view using Quartus:

```
1  module Verilog1
2  (
3      input  wire A,
4      input  wire B,
5      input  wire Cin,
6      output wire Sum,
7      output wire Cout
8  );
9
10     assign Sum  = A ^ B ^ Cin;
11     assign Cout = (A & B) | (Cin & (A ^ B));
12
13 endmodule
14
15
```

